Kennedy
Baptist College

Semester Two Examination, 2019

Question/Answer booklet

MATHEMATICS APPLICATIONS UNITS 1 AND 2
Section One: Calculator-free

SOLUTIONS

Student number: In figures

In words ______________________

Your name ______________________

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Section One: Calculator-free 35% (52 Marks)

This section has **eight (8)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1 **(6 marks)**

(a) Simplify $2 + 4 \times \sqrt{36} - 20 \div 2^2$. (2 marks)

Solution
$2 + 4 \times \sqrt{36} - 20 \div 2^2 = 2 + 4 \times 6 - 20 \div 4$ $= 2 + 24 - 5$ $= 21$
Specific behaviours
✓ indicates several correct steps   ✓ correct value

(b) Determine the value of the expression $\dfrac{x^2 + 2x}{y^2 - 3y}$ when $x = 5$ and $y = -2$. (2 marks)

Solution
$\dfrac{5^2 + 2(5)}{(-2)^2 - 3(-2)} = \dfrac{25 + 10}{4 + 6}$ $= \dfrac{35}{10}$ $= 3.5$
Specific behaviours
✓ indicates several correct steps   ✓ correct value

(c) Determine the value of d when $a = 2.8, v = 0.5$ and $d = 6v^2 - 2av$. (2 marks)

Solution
$d = 6 \times 0.5 \times 0.5 - 2 \times 2.8 \times 0.5$ $= 3 \times 0.5 - 2.8$ $= 1.5 - 2.8$ $= -1.3$
Specific behaviours
✓ indicates several correct steps   ✓ correct value

Question 2 **(6 marks)**

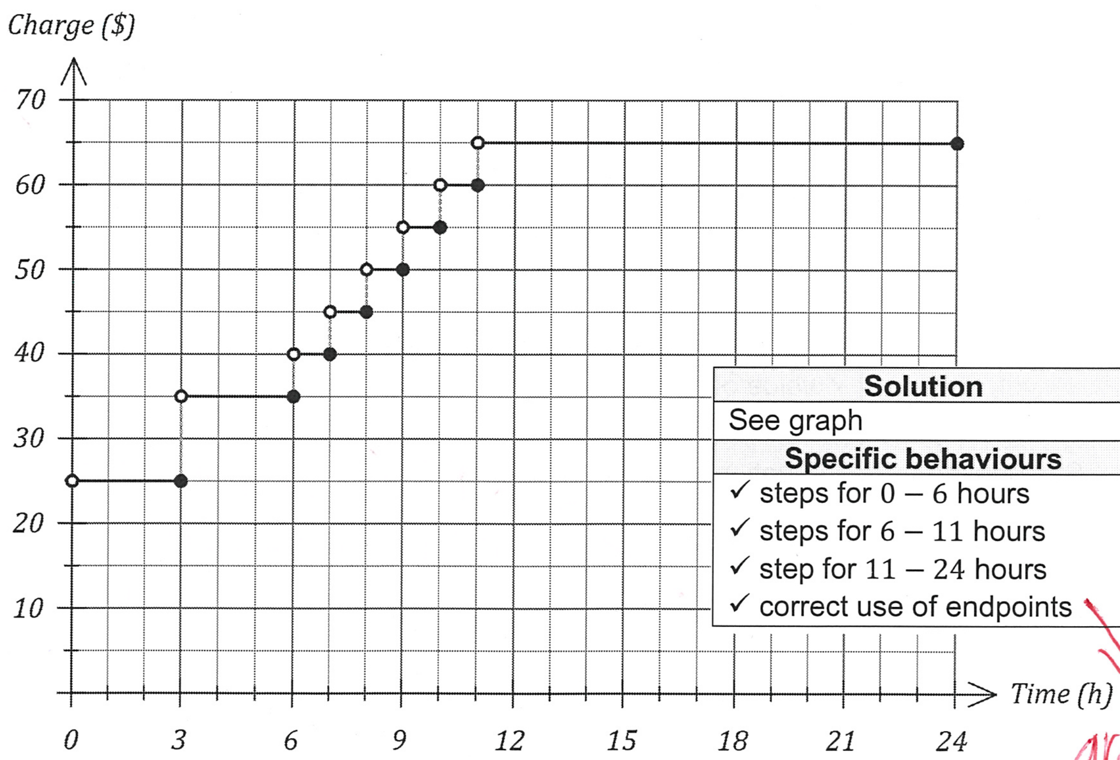
An airport car park charges the amounts shown in the table below for periods not exceeding 24 hours.

Parking time	No more than 3 hours	More than 3 but not exceeding 6 hours	Each additional hour (or part) exceeding 6	Maximum charge for up to 24 hours
Charge	$25	$35	$5	$65

(a) Explain why the charge to park a car for eight and a half hours is $50. (2 marks)

Solution
Must pay for 6 hours plus additional 3 hours: $$C = 35 + 3 \times 5 = 35 + 15 = \$50$$
Specific behaviours
✓ indicates 3 additional hours ✓ shows calculation

(b) Draw a graph to represent the parking charges on the axes below. (4 marks)



Solution
See graph
Specific behaviours
✓ steps for 0 – 6 hours ✓ steps for 6 – 11 hours ✓ step for 11 – 24 hours ✓ correct use of endpoints

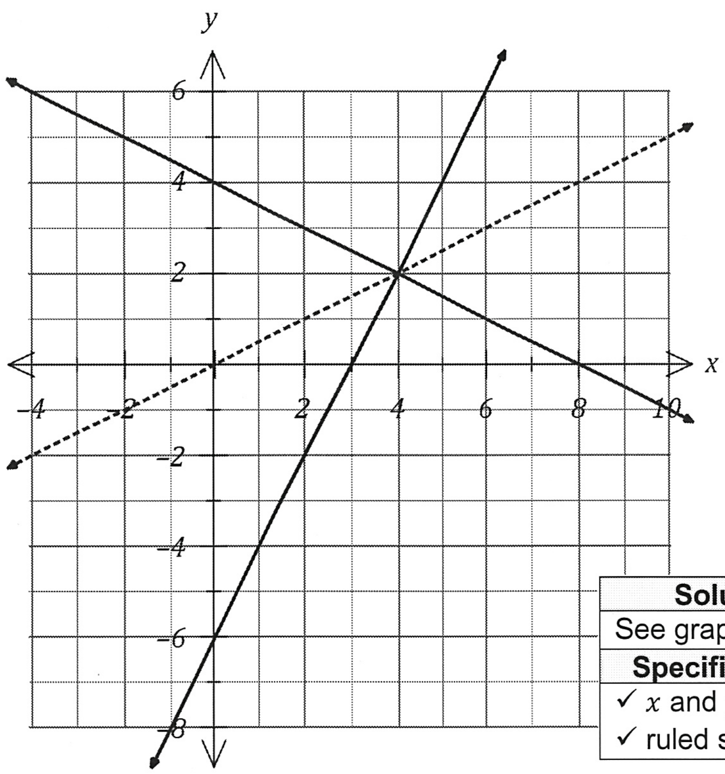
Only if graph has been attempted completely.

-1 mark if lines are poorly sketched/ruled improperly

Question 3 **(7 marks)**

The graph of line L_1 with equation $y = ax + b$ is shown below.

* Be lenient on drawing of lines due to poor print quality *



Solution (b)(i)
See graph
Specific behaviours
✓ x and y intercepts
✓ ruled straight line

Must have one correct intercept for mark.

(a) Determine the value of the constant a and the value of the constant b. (2 marks)

Solution
Gradient $= a = 2$. y-intercept $= b = -6$.
Specific behaviours
✓ value of a
✓ value of b

(b) Straight-line L_2 has equation $x + 2y = 8$.

No F.T from diagram unless a wrong equation was clearly shown.

(i) Draw L_2 on the axes above. (2 marks)

(ii) State the gradient of L_2. (1 mark)

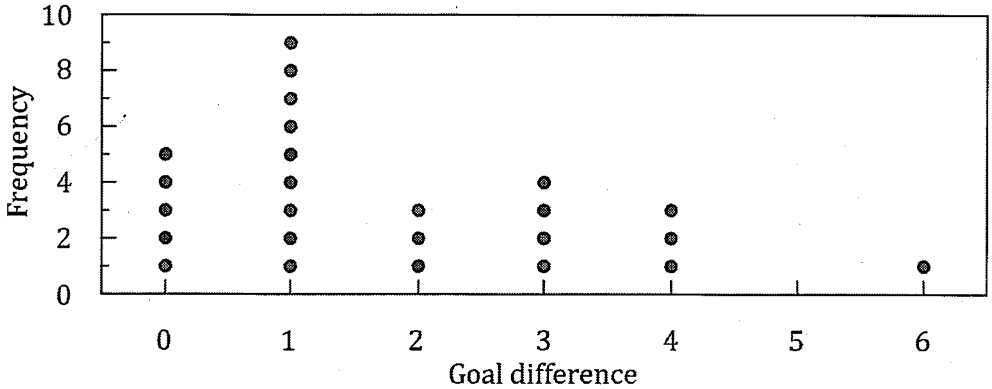
Solution
$m = -\dfrac{1}{2} = -0.5$
Specific behaviours
✓ correct gradient

(c) Determine the equation of the straight-line L_3 that passes through the origin and the point of intersection of lines L_1 and L_2. (2 marks)

Solution
$m = \dfrac{2}{4} = \dfrac{1}{2} \Rightarrow y = \dfrac{1}{2}x$
Specific behaviours
✓ indicates correct gradient
✓ correct equation

Question 4 **(9 marks)**

The dot plot below shows the goal difference between each pair of teams playing in a weekend round of a soccer league.



(a) State the number of games that did not end in a draw. (1 mark)

Solution
$25 - 5 = 20$ games
Specific behaviours
✓ correct number

(b) Classify the type of variable that goal difference is by circling two of the following words: (1 mark)

Categorical. Continuous. Discrete. Nominal. Numerical. Ordinal.

Solution
Discrete and Numerical
Specific behaviours
✓ clearly circles both

(c) Determine

(i) the median goal difference. (1 mark)

Solution
Median is 13^{th} score $= 1$
Specific behaviours
✓ correct median

(ii) the mean goal difference. (2 marks)

Solution
$\dfrac{0+9+6+12+12+6}{25} = \dfrac{45}{25} = \dfrac{9}{5} = 1.8$
Specific behaviours
✓ indicates correct method
✓ correct mean (as fraction or decimal)

(d) Describe the distribution of the dataset in terms of modality, shape and location. (3 marks)

Solution
The dataset has one peak and hence is unimodal. The shape of the dataset displays positive skew. The dataset is located around the mean of 1.8 goals.
Specific behaviours
✓ indicates unimodal ✓ indicates positive skew ✓ uses mean or median or mode for location

Pay mark if mean, median or mode stated.

(e) Describe the feature of the dot plot that would suggest the median would be less than the mean of the dataset. (1 mark)

Solution
The positive skew.
Specific behaviours
✓ answer includes positive skew

Can describe the skew in detail instead

i.e "The data was skewed to the right side"

Question 5 **(7 marks)**

The number of minutes that a group of students took to complete a task are listed below:

$$29, 26, 17, 29, 18, 26, 31, 25, 26, 35, 33, 28$$

(a) Construct an ordered stem-and-leaf plot to display this data. (3 marks)

Solution
$\begin{array}{r\|l} 1 & 7\ 8 \\ 2 & 5\ 6\ 6\ 6\ 8\ 9\ 9 \\ 3 & 1\ 3\ 5 \end{array}$ Key 2|5 = 25
Specific behaviours
✓ stems and leaves aligned in columns and key shown ✓ two correct set of leaves ✓ all sets of leaves, correctly ordered

(b) Identify, with justification, any outliers that may be present in the times. (4 marks)

Solution
$Q_1 = 25.5, \qquad Q_3 = 30$ $1.5 \times IQR = 1.5 \times 4.5 = 6.75$ $Q_1 - 6.75 = 25.5 - 6.75 = 18.75$ Any times less than 18.75 minutes are outliers. Hence the times of 17 and 18 minutes are both outliers.
Specific behaviours
✓ identifies quartiles ✓ calculates $1.5IQR$ ✓ calculates lower bound ✓ identifies both outliers

Question 6 **(5 marks)**

(a) Solve the equation $3(x+5)-1=x-2(x-3)$. (2 marks)

Solution
$3x+15-1=x-2x+6$ $4x=-8$ $x=-2$
Specific behaviours
✓ correctly expands ✓ correct solution

(b) Use the formula $A=\frac{1}{2}(a+b)h$ to determine

(i) A when $a=5.6, b=8.4$ and $h=20$. (1 mark)

Solution
$A=\frac{1}{2}(5.6+8.4)\times 20$ $=10\times 14$ $=140$
Specific behaviours
✓ correct value

(ii) b when $A=33, a=4$ and $h=6$. (2 marks)

Solution
$33=\frac{1}{2}(4+b)\times 6$ $33=3(4+b)$ $11=4+b$ $b=7$
Specific behaviours
✓ substitutes and simplifies ✓ correct value of b

Question 7 **(6 marks)**

(a) Simplify $2\begin{bmatrix} 5 & 2 \\ -3 & 4 \end{bmatrix} - \begin{bmatrix} 6 & -1 \\ 2 & 7 \end{bmatrix}$. (2 marks)

Solution
$\begin{bmatrix} 10 & 4 \\ -6 & 8 \end{bmatrix} - \begin{bmatrix} 6 & -1 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ -8 & 1 \end{bmatrix}$
Specific behaviours
✓ correct multiple ✓ correct matrix

(b) Given that $A = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$ determine A^2. (2 marks)

Solution
$\begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$
Specific behaviours
✓ at least two elements correct ✓ correct matrix

(c) Determine the value of x and the value of y given that $\begin{bmatrix} 2 \\ 5 \end{bmatrix} + x\begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} -10 \\ 3y \end{bmatrix}$. (2 marks)

Solution
$2 - 3x = -10 \Rightarrow x = 4$ $5 + 4(4) = 3y \Rightarrow y = 7$
Specific behaviours
✓ value of x ✓ value of y

Question 8 **(6 marks)**

At the start of the year 2002, the difference in the ages of a mother and her child was 24.

Let the age of the mother be x and the age of her child be y at the start of 2002.

(a) Use the above information to write an equation relating x and y. (1 mark)

Solution
$x - y = 24$
Specific behaviours
✓ correct equation

or $y = x - 24$
or $x = y + 24$

Three years later, the mother was 5 times as old as her child.

(b) Use this additional information to write another equation relating x and y. (2 marks)

Solution
In 3 years, mother will be $x+3$ and child will be $y+3$. $x + 3 = 5(y + 3)$
Specific behaviours
✓ expressions for ages of mother and child in 3 years   ✓ correct equation

(c) Determine the age of the child at the start of the year 2019. (3 marks)

Solution
Substitute $x = y + 24$ into second equation: $y + 24 + 3 = 5y + 15$ $4y = 12 \Rightarrow y = 3$ In 2019 child will be $y + 17$ and so child will be 20 years old.
Specific behaviours
✓ substitutes and expands   ✓ solves for y ✓ correct age of child

ft from (b)
If $x = 5y$
then $y = 6$
$\Rightarrow$ child is 23

End of questions